



How Youth-Led Community Mapping *Boosted Climate Preparedness in Timor-Leste*

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SUMMARY

In municipalities like Bobonaro, in Timor-Leste, communities face rising threats from flooding and drought, worsened by climate change. Yet, until recently, disaster risk planning relied on hand-drawn maps and fragmented information. HOT's Asia Pacific Open Mapping Hub and World Vision Timor-Leste (WVTL) together with the youth group, Grupo Sistema Informasaun Geografia (G-SIG), trained local volunteers to collect and visualize risk data using open-source tools. These efforts not only filled critical data gaps but also helped shape decisions—from identifying the most at-risk people and designing evacuation routes to improving access to water in drought-prone villages.

UNDERSTANDING THE CONTEXT

It starts with a basic question: **how do you prepare a village for a disaster you can't see coming?**

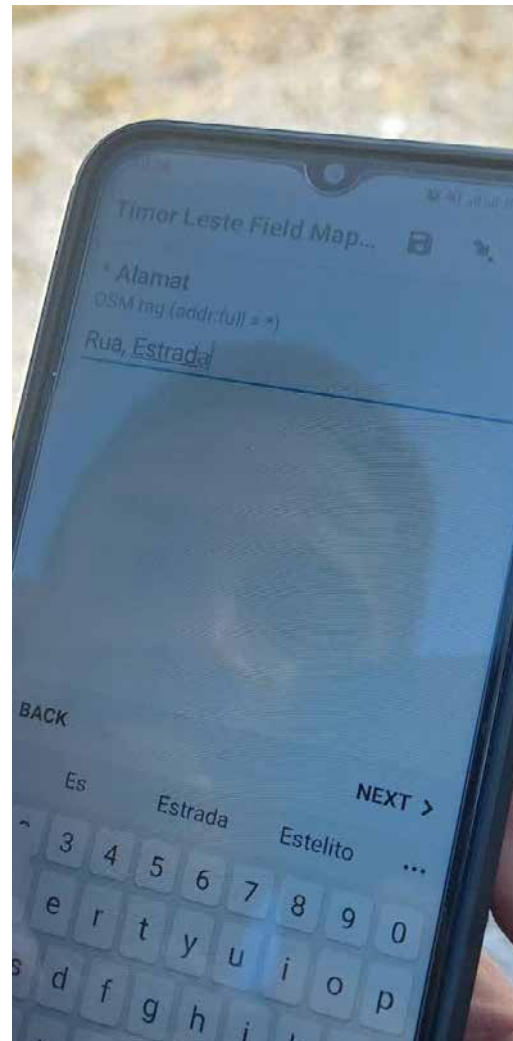
For years, staff at World Vision Timor-Leste (WVTL) struggled to answer this with the limited tools they had. Risk assessments relied on maps drawn by hand on flipcharts, passed from one field team to the next, at times getting damaged. There was no reliable system to show where the flood waters might rise, or which homes would run dry in a drought.

Meanwhile, the climate crisis deepened. Timor-Leste began facing stronger and more frequent extreme weather events—flash floods that swept away crops, and longer dry seasons that left families without water ([European Commission's Disaster Risk Management Knowledge Centre, n.d.](#); [World Bank Group and Asian Development Bank, 2021](#)). **Communities in Bobonaro, one of the country's climate-vulnerable municipalities, were among the hardest hit.**

¹ All research, analysis, and conclusions are the authors' own. OpenAI was used to assist with drafting, and the text was reviewed, edited and validated by humans.

World Vision had a growing number of disaster risk management and anticipatory action projects in Timor-Leste. Yet, back in 2021, WVTL faced major constraints. Zito Soares, previous Disaster Ready Project Coordinator, and now the lead of the Humanitarian and Emergency Affairs of WVTL, shared how their risk assessments previously depended on hand-drawn maps, often lost or damaged during field visits. This made it difficult to compile consistent disaster data across project areas.

Then, in 2022, as part of the broader Disaster READY Timor-Leste initiative under the Australian Humanitarian Partnership (AHP), WVTL teamed up with HOT's Asia Pacific Open Mapping Hub and Grupo Sistema Informasaun Geografia (G-SIG) to prepare for the implementation of Phase II of the project, which was looking to enhance disaster preparedness across municipalities. **Their goal: generating reliable geospatial data to better inform local disaster preparedness, led by local actors on the ground** (Australian Humanitarian Partnership, n.d.).



Field Mapping in Baucau Municipality (WVTL, 2022).

OUR APPROACH IN ACTION

Between July and October 2022, HOT and WVTL trained over 30 participants—including nine women—from NGOs, government offices, universities, and social enterprises. Participants learned how to use OpenStreetMap (OSM) for data digitization, field data collection (including mobile mapping with KoboToolbox), data analysis and risk modeling.

With guidance from HOT and WVTL, the youth-led GIS group, G-SIG, played a key role in leading field mapping in Bobonaro. Using tools like the HOT Tasking Manager, HOT Export Tool, and InaSAFE, local

teams engaged community members in mapping **more than 47,000 buildings, 350 kilometers of roads, and over 700 community facilities across Bobonaro**, as well as two other municipalities: Baucau and Aileu.

This data was uploaded to OSM and cross-referenced with known hazard zones.

For the first time, WVTL had a spatial picture of where climate risks were most concentrated.

Open Mapping Training with Gov't, NGO, and community members
(WVTL, 2022)



IMPACT ON PEOPLE AND COMMUNITIES

With this new data, WVTL and local governments shaped their anticipatory action efforts. In the Bobonaro municipality, **hazard maps helped identify villages most at risk of flooding.** These maps guided early warning efforts and informed evacuation planning for at least 80

households in sub-villages Ilat-Laun and Suriubu of Bobonaro (World Vision Timor-Leste, 2024). As residents were engaged in mapping, **Julmira de Almeida, Project Lead of G-SIG,** noted that:

“when disasters happen, [community members] know the place that’s safe for them to go and what are things that they must do to protect themselves during disasters.”

WVTL also used the skills and knowledge gained through HOT’s training to continue community mapping activities that inform drought response.

Baucau. With it, they can report on soil and water conditions, helping build long-term climate resilience of 15 Farming Land Sites in two villages in Baucau.

Together, WVTL, G-SIG, and community members co-created an El Niño Monitoring Report that mapped and identified areas facing severe water shortages. In eight drought-affected villages, WVTL installed 15,000-liter water tanks, refilled daily, ensuring about 438 households had access to water through the dry season (World Vision Timor-Leste, 2024).

Finally, under WVTL’s Regreening Communities Project, young farmers are learning to map and use a Regreening Monitoring Platform to track environmental restoration of their fields in Bobonaro and

“This data really helps us —not only for disaster preparedness planning, but for all projects, whether it’s on climate resilience, child protection, or wider development work. Before starting interventions, we now better understand the situation of communities in vulnerable areas.”

—Zito Afranio Soares,
Humanitarian and Emergency
Affairs Manager, WVTL

HOW CHANGE CAN LAST

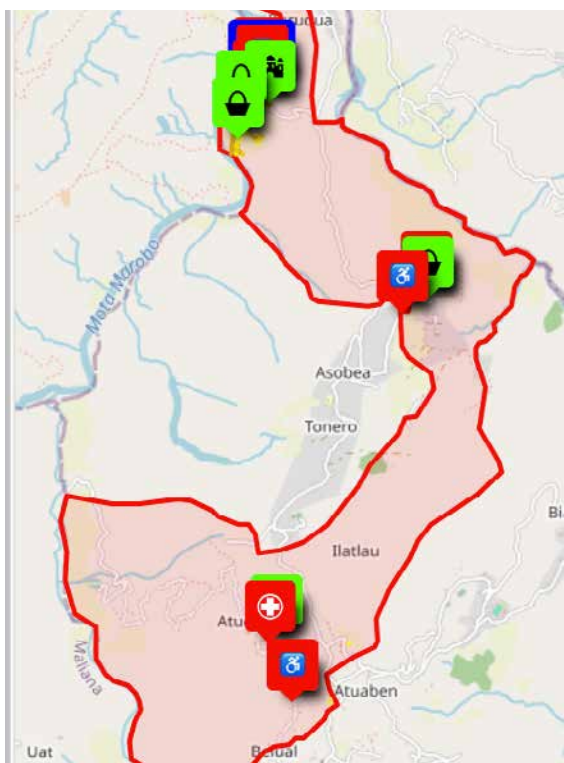
This initiative sparked the formation of the OSM Timor-Leste Community, a network of volunteer mappers from government, civil society, and youth groups.

Beyond the partnership with HOT, WVTL invested in strengthening G-SIG's role as youth-led mapping organization—supporting its organizational capacity development to lead mapping initiatives across sectors. This extended beyond the Disaster Ready Program, as WVTL continued to work with G-SIG to better engage the youth and community members in planning.

“After we’re involved in open mapping, it’s like a dream come true.

A lot of agencies want to collaborate, [they] want us to, help them in mapping, provide them not only data, but also provide them skills in mapping using open mapping software such as QGIS, Ushahidi, and OpenStreetMap.”

Ponciano da Costa de Jesus,
Executive President, G-SIG



Screenshot of WVTL and G-SIG's Anticipatory Action Flood Risk Map of Ilat-Laun, Bobonaro, visualized through [uMap](#).

Zito Soares, Humanitarian and Emergency Affairs Manager of WVTL, noted that they continue their collaboration to map and identify households living in vulnerable areas, aiding WVTL in providing support where it is most needed (World Vision Timor-Leste, personal communication, May 2, 2025).

WVTL's example has inspired others in the Australian Humanitarian Partnership network to explore how open data can improve their work towards inclusive community-based disaster risk management and anticipatory action. The Timor-Leste government is now looking at ways to formally integrate OSM data into disaster workflows.

Meanwhile, community members who once relied on flipcharts are building the digital maps that guide their own resilience.

INSIGHTS FOR THE FUTURE

This project shows how local organizations can engage youth groups and community members with open geospatial data and tools to fill critical gaps in areas vulnerable to disaster. What started as a participatory mapping effort has evolved into a movement that has proven effective in preparing communities before climate disasters strike, taking active roles in local climate action, benefiting from early warning systems for flood-prone villages, increased water access for drought affected communities and soil monitoring for farming.



Field Mapping in Baucau Municipality (WVTL, 2022).

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